TABLE II
BENCHMARK RESULTS ACROSS EIGHT TEST CONDITIONS

Method	Train ep.	J (%) ↑ [95% CI]	L_1 (cm) ↓	L_2 (cm) ↓	L_3 (%) ↓
Reference rows, shared MPPI					
U (oracle)	–	76.3 [65.0, 86.8]	0	0	0
Physics + mass est.	–	77.8 [67.5, 87.0]	0.57	0.47	3.7
Physics (25 g)	–	53.3 [42.5, 64.3]	8.42	3.62	33.8
Learned world models, shared MPPI					
Example MLP	–	15.3 [9.3, 22.3]	12.81	4.67	51.9
JEPA `full`	500	34.9 [26.5, 43.6]	10.84	4.07	47.8
JEPA `full`	2,000	50.4 [40.3, 60.3]	9.01	3.56	47.6
JEPA `no_cross`	500	30.0 [22.6, 37.8]	10.67	3.82	46.7
JEPA `no_cross`	2,000	49.4 [39.8, 59.0]	8.80	3.48	46.1
JEPA `flat`	500	12.2 [7.1, 18.3]	12.59	4.41	51.4
JEPA `flat`	2,000	16.2 [11.4, 21.5]	11.20	4.25	56.4
W-ar	500	45.3 [36.2, 54.3]	9.46	3.44	43.7
W-ar	2,000	46.8 [37.8, 55.5]	8.55	3.07	41.4
PETS	500	17.2 [11.4, 23.7]	13.24	3.92	47.8
PETS	2,000	23.3 [16.4, 30.9]	12.53	3.72	46.6
Policies without the shared planner					
CrazyMARL	–	40.6 [30.0, 51.5]	–	–	–
TD-MPC2	–	0.0[a]	–	–	–
DreamerV3	–	0.0[a]	–	–	–

400 test episodes per model. For three-seed methods, J pools 1,200 episodes; L_1–L_3 are seed means. Brackets: 95% bootstrap CIs over 50 layouts. Train ep.: offline training episodes; –: not reported or not applicable. [a]Zero successes across all three seeds.